

Reproducibility Engineering in Data Management Research

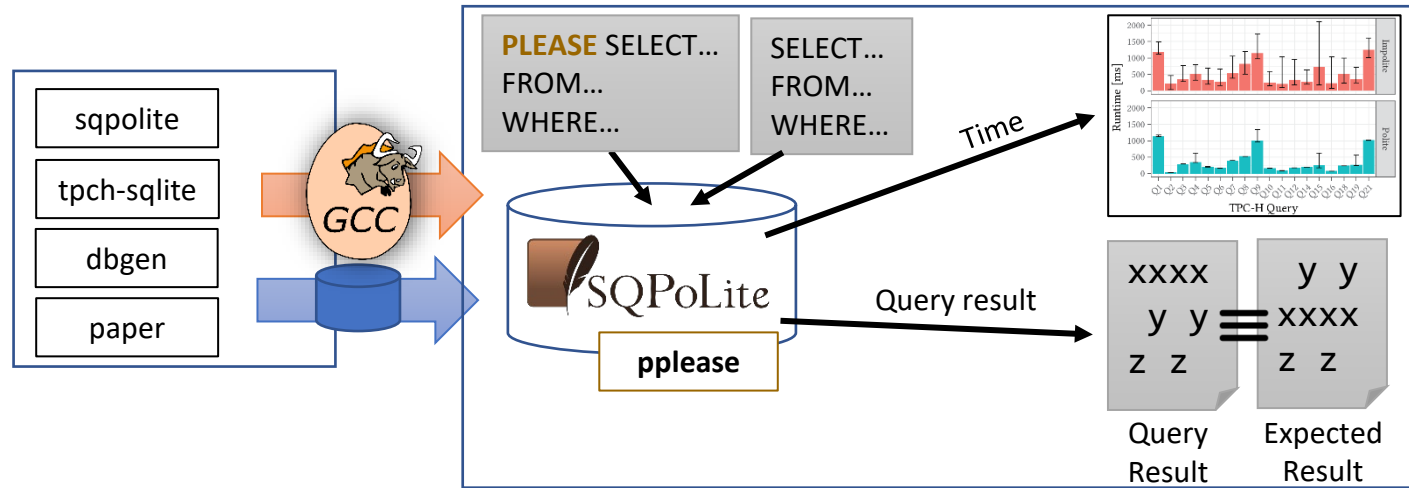


<https://opendreamkit.org/2017/11/02/use-case-publishing-reproducible-notebooks/>

Stefanie Scherzinger

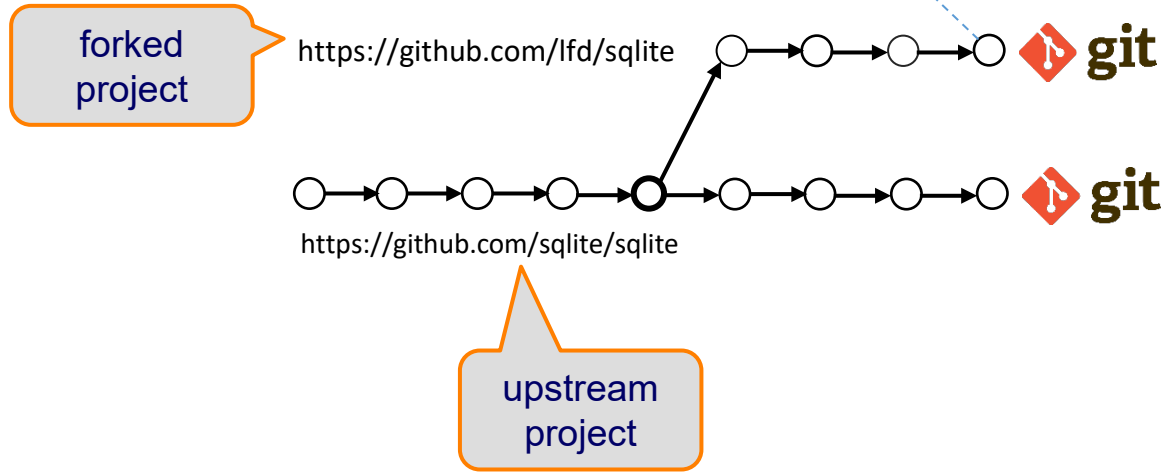
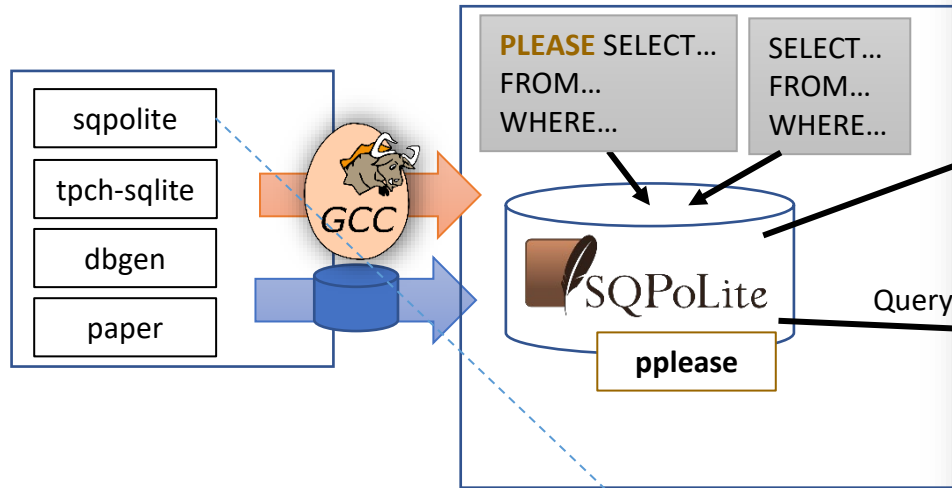
Material includes joint work with Wolfgang Mauerer (OTH Regensburg):

- ICDE 2021 tutorial “Nullius in Verba: Reproducibility for Database Systems Research, Revisited.” (Mauerer, Scherzinger)
- Flipped classroom course on “Reproducibility Engineering”, jointly taught at OTH Regensburg (Mauerer) and Uni Passau (Scherzinger, Klessinger) (videos: <https://tinyurl.com/repeng>)
- SEENG@ICSE 2022 paper “Beyond the Badge: Reproducibility Engineering as a Lifetime Skill.” (Mauerer, Klessinger, Scherzinger)



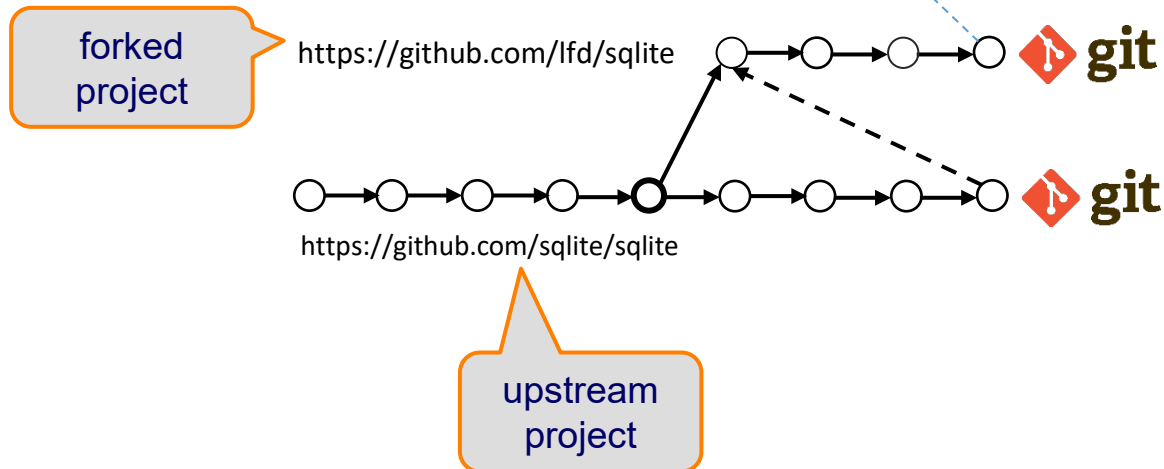
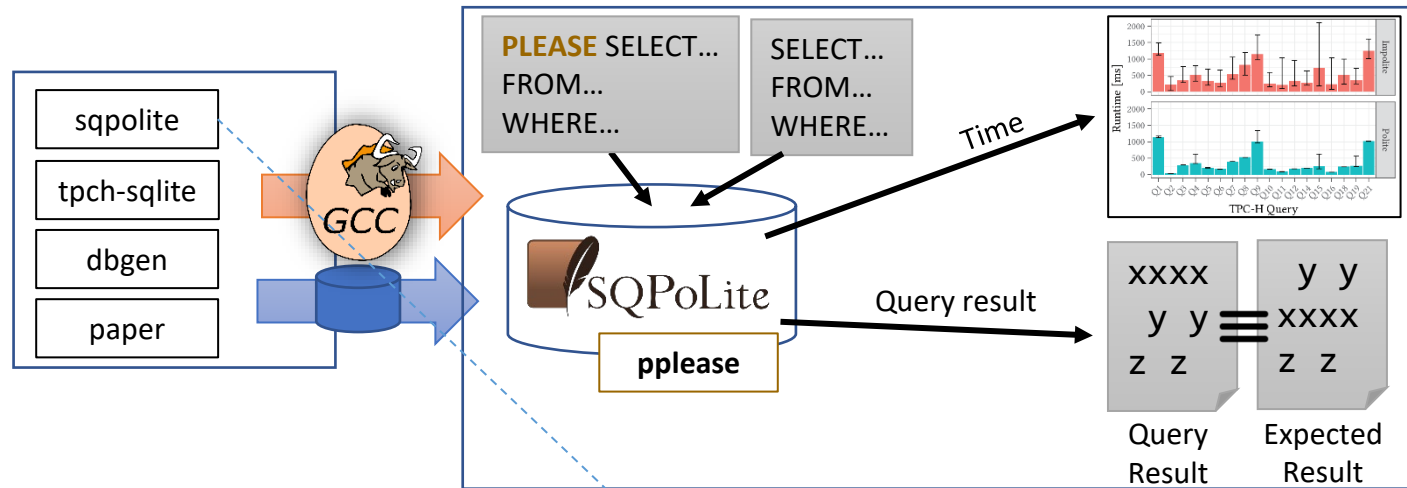
Brief Demo

Mini Tutorial: Packaging Artefacts



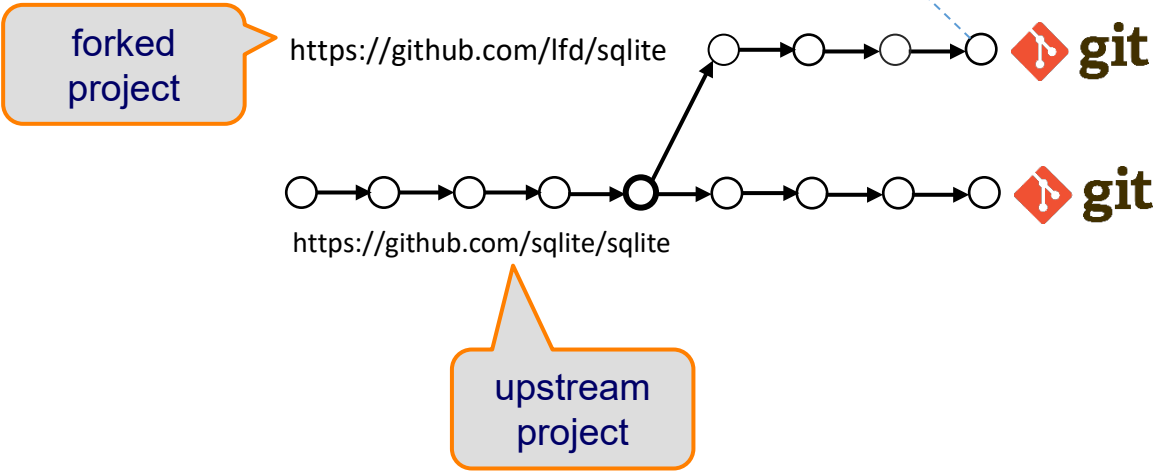
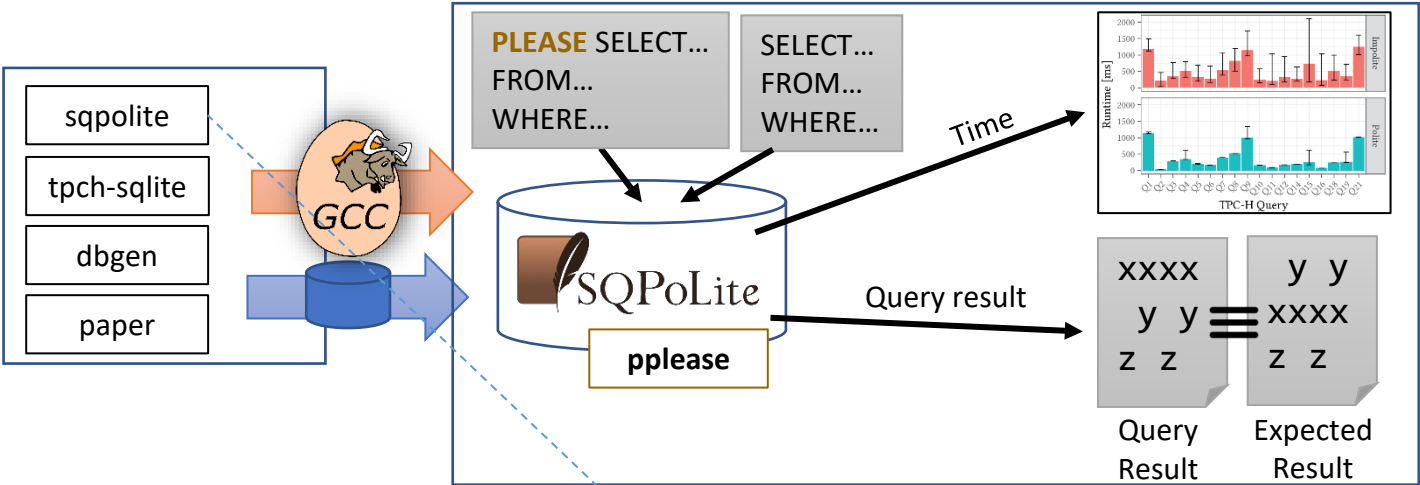
The screenshot shows the GitHub repository page for `lfd/sqlite`. The repository is public and has 3 watchers, 1 fork, and 0 stars. The main branch is `master`. The commit history is displayed, showing commits from June 14, 2022, and March 28, 2021. The commits include:

- Add latency measurement code** (commit `03592f2`) by Wolfgang Maurer and ase-double-blind.
- Punish naughty users with sleep** (commit `d74de15`) by Wolfgang Maurer and ase-double-blind.
- Ensure bidirectional friendliness** (commit `0387564`) by Wolfgang Maurer and ase-double-blind.
- Try to please everyone by being polite** (commit `be89a8b`) by Wolfgang Maurer and ase-double-blind.
- Properly handle folders containing whitespace during configure** (commit `e6279a1`) by Wolfgang Maurer.
- Document the "%token" directive for Lemon. This directive has been in...** (commit `9cffb0f`) by drh.
- For the sqlite3_bind_text16 TCL binding used for testing, ensure that...** (commit `5e23ae5`) by drh.
- In the sqlite3_bind_text and sqlite3_bind_text16 TCL commands of the ...** (commit `639ae98`) by drh.



Fork inherits commit history of the upstream project, and preserves the attribution.

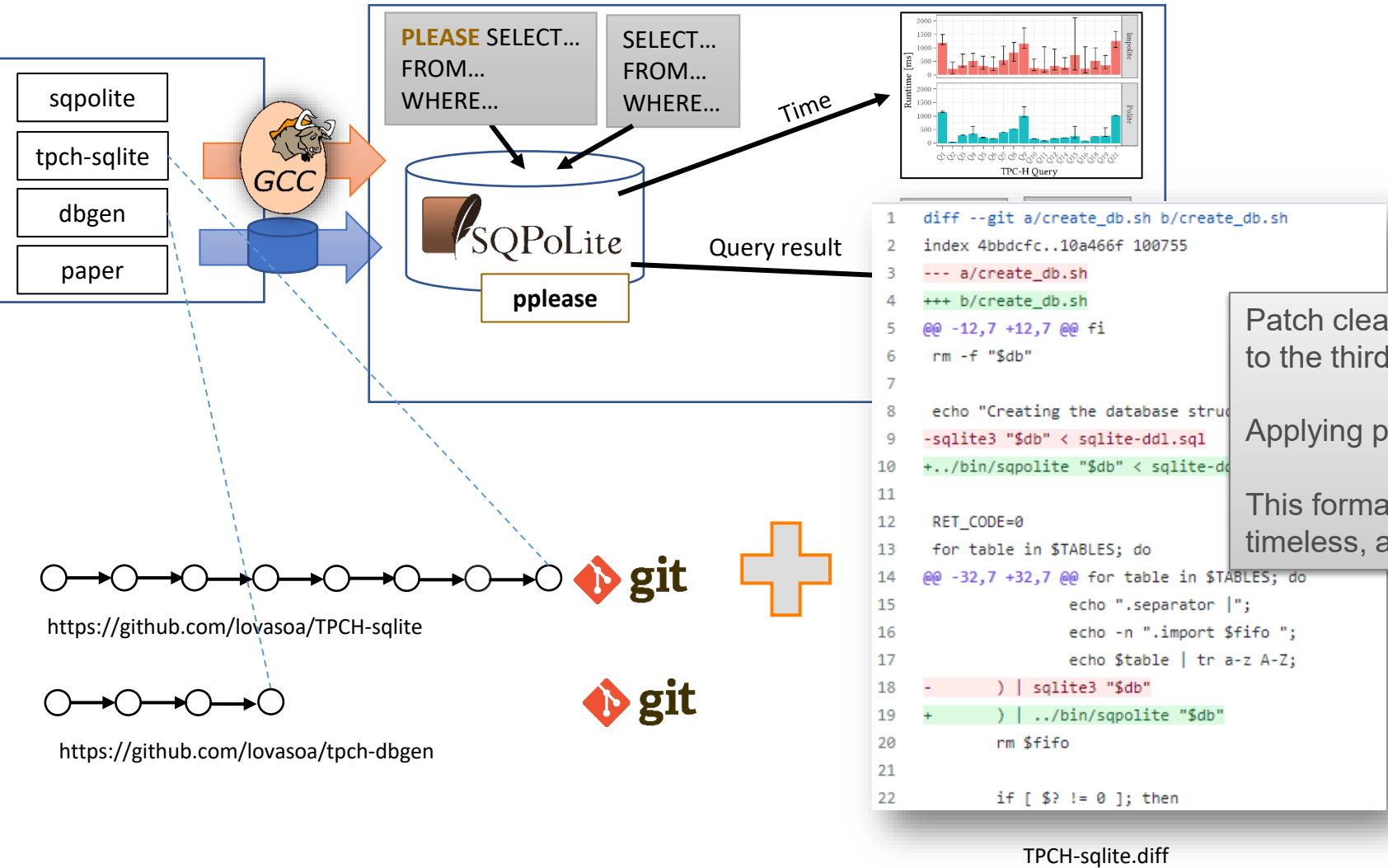
Later, we may even rebase our changes onto a newer commit in the upstream project.



```
WORKDIR /home/repro/git-repos
RUN git clone https://github.com/lfd/sqlite.git
```



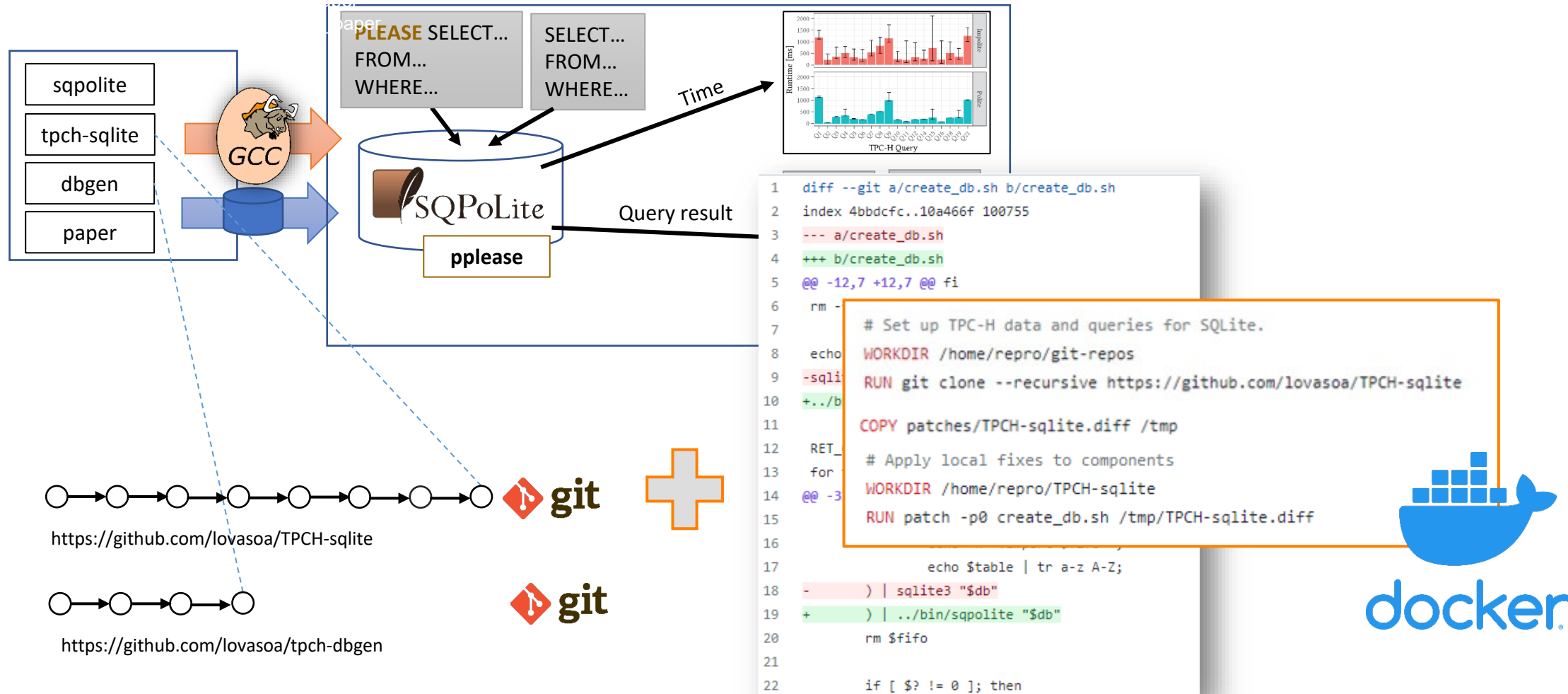
docker

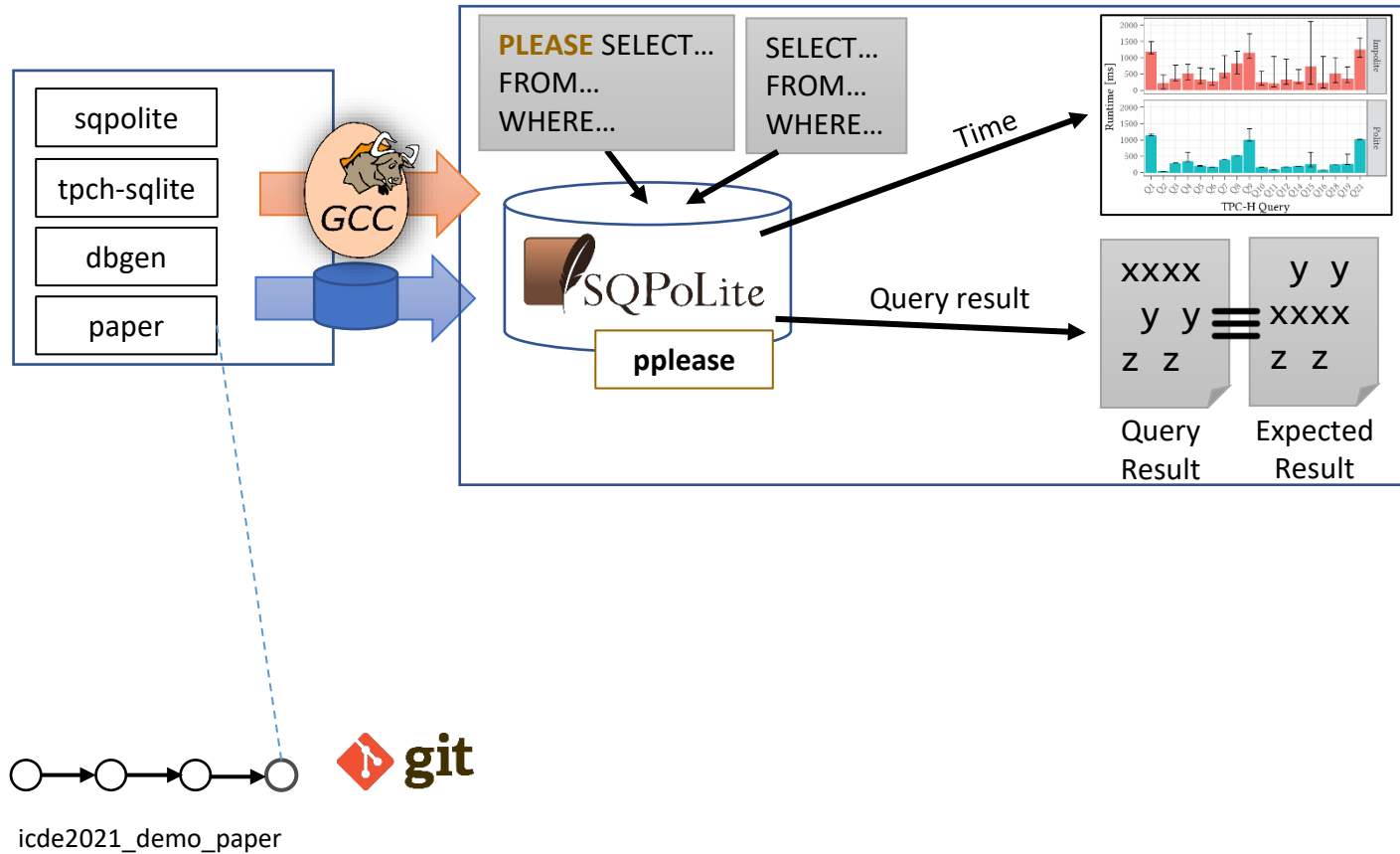


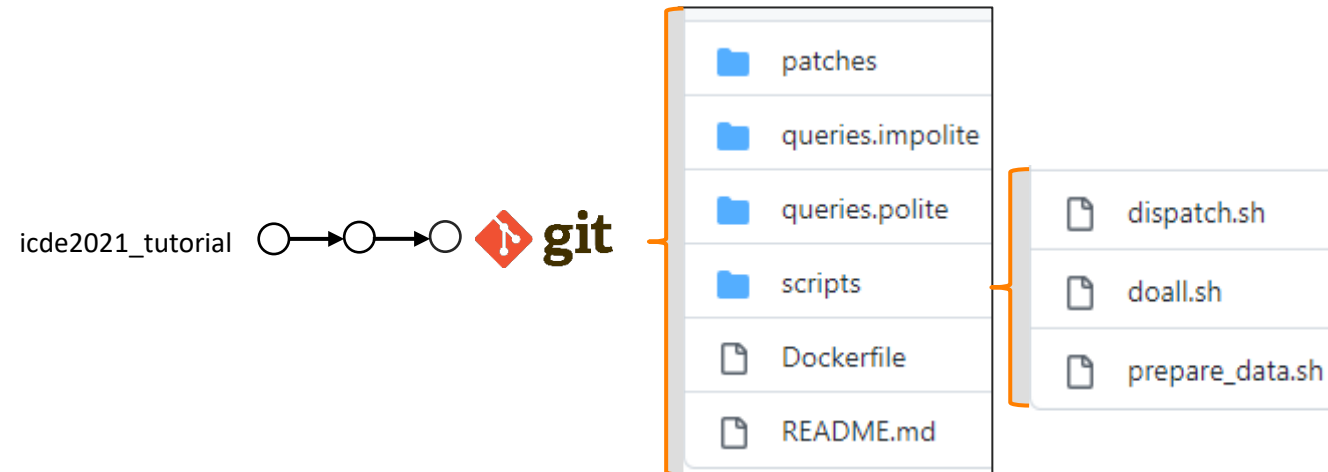
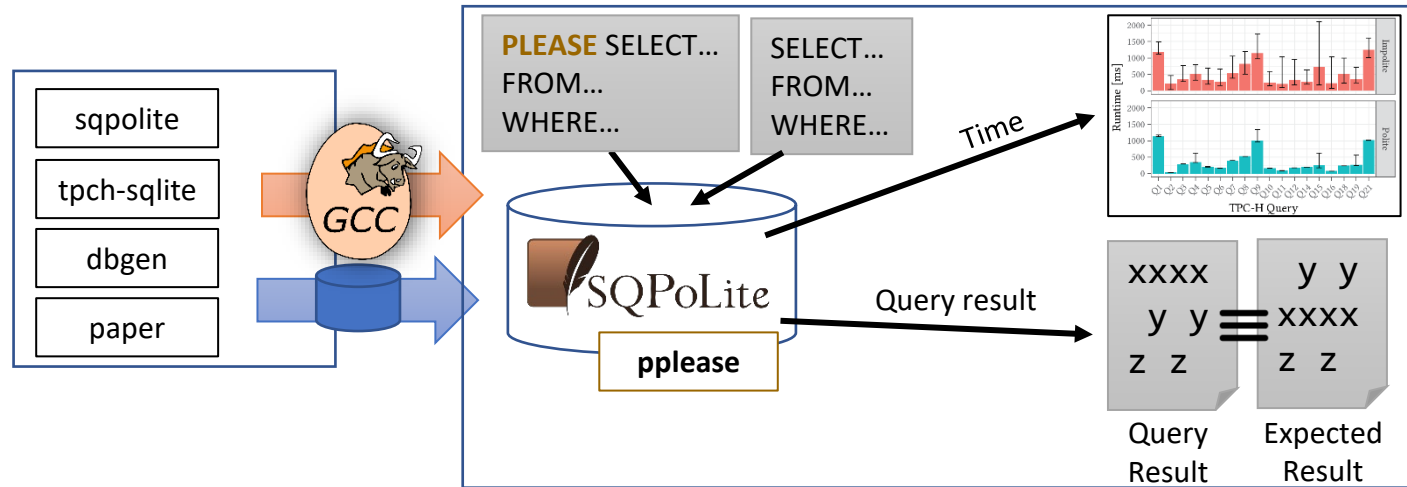
Patch clearly documents our changes to the third-party code.

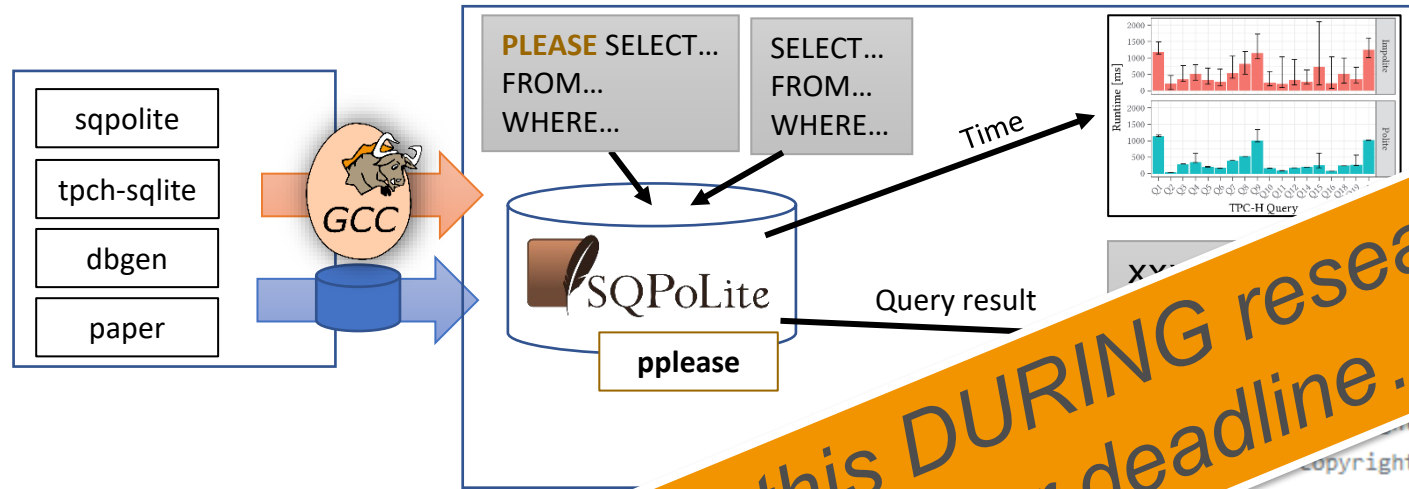
Applying patches can be automated.

This format is very simple, stable, timeless, and does not even rely on *git*.

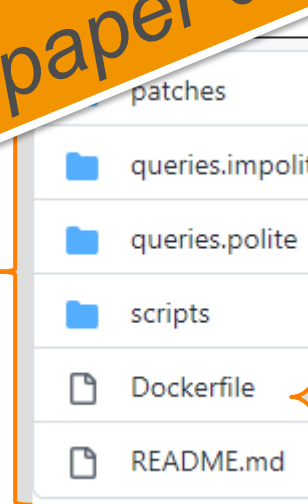




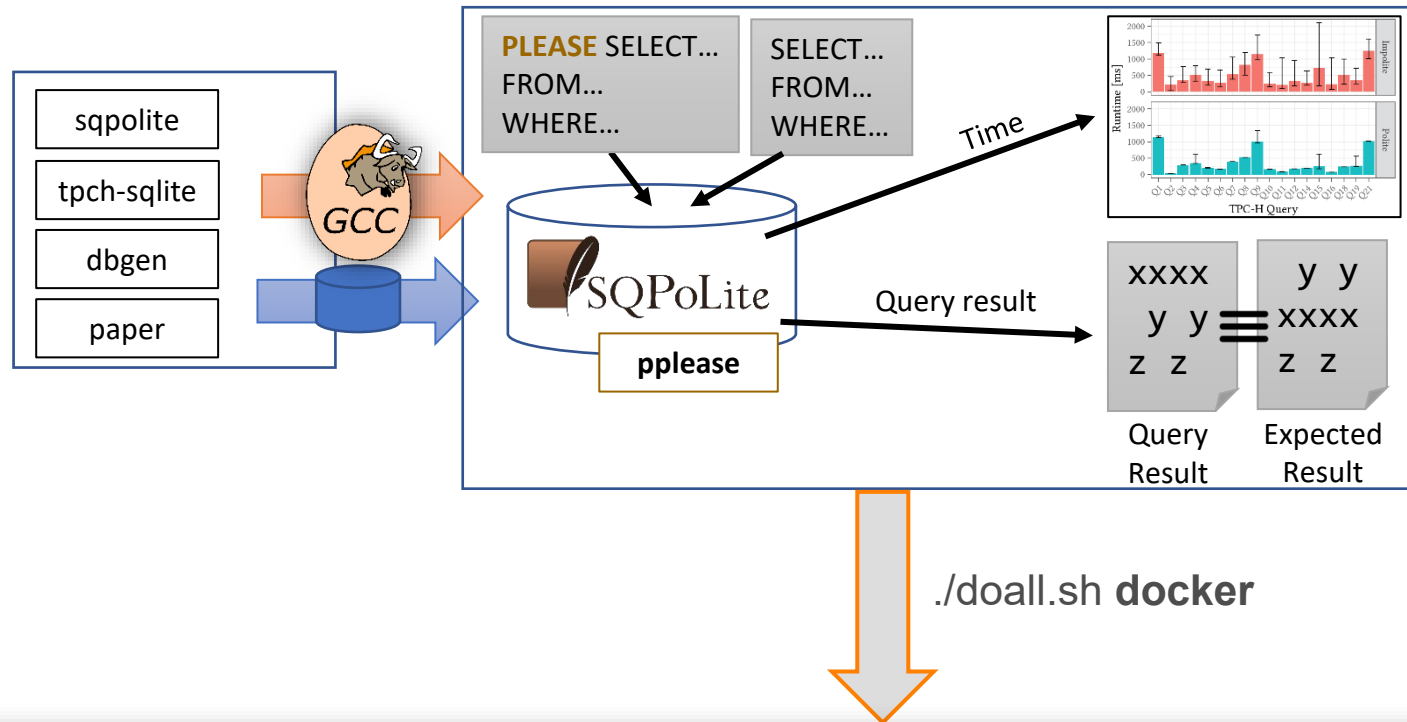




Ideally, you do this **DURING** research,
not after the paper deadline...



for ICDE 2021 tutorial
ducibility for Database Systems Research, Revi
Copyright 2021, Wolfgang Mauerer <wolfgang.mauerer@oethr.de>
Copyright 2021, Stefanie Scherzinger <stefanie.scherzinger@uni-passau.de>
SPDX-License-Identifier: GPL-2.0-only
Start off of a long-term maintained base distribution
FROM ubuntu:18.04
MAINTAINER Wolfgang Mauerer <wolfgang.mauerer@oethr.de>
MAINTAINER Stefanie Scherzinger <stefanie.scherzinger@uni-passau.de>
ENV DEBIAN_FRONTEND noninteractive
ENV LANG="C.UTF-8"
ENV LC_ALL="C.UTF-8"
RUN apt update -qq
RUN apt-get install -y gpg-agent
RUN apt install -y --no-install-recommends software-properties-common di
RUN apt-key adv --keyserver keyserver.ubuntu.com --recv-keys E298A3A825C
RUN add-apt-repository "deb https://cloud.r-project.org/bin/linux/ubuntu
RUN add-apt-repository ppa:c2d4u.team/c2d4u4.0+



```
repro@ec0cff2aa4e8: ~  
repro@ec0cff2aa4e8:~$ ls -l results/  
total 8  
drwxr-xr-x 3 repro repro 4096 Aug 23 18:07 res_SF-0.1_scenario-impolite_docker  
drwxr-xr-x 3 repro repro 4096 Aug 23 18:05 res_SF-0.1_scenario-polite_docker
```

COMMENT | FOCUS

Check for updates

Reproducibility standards for machine learning in the life sciences

To make machine-learning analyses in the life sciences more computationally reproducible, we propose standards based on data, model and code publication, programming best practices and workflow automation. By meeting these standards, the community of researchers applying machine-learning methods in the life sciences can ensure that their analyses are worthy of trust.

Benjamin J. Heil, Michael M. Hoffman, Florian Markowitz, Su-In Lee, Casey S. Greene and Stephanie C. Hicks

The field of machine learning has grown tremendously within the past ten years. In the life sciences, machine-learning models are rapidly being adopted because they are well suited to cope with the scale and complexity of biological data. However, there are drawbacks to using such models. For example, machine-learning models can be harder to interpret than simpler models, and this opacity can obscure learned biases. If we are going to use such models in the life sciences, we will need to trust them. Ultimately all science requires trust —no scientist can reproduce the results from every paper they read. The question, then, is how to ensure that machine-learning analyses in the life sciences can be trusted.

One attempt at creating trustworthy analyses with machine-learning models revolves around reporting analysis details such as hyperparameter values, model architectures and data-splitting procedures. Unfortunately, such reporting requirements are insufficient to make analyses trustworthy. Documenting implementation details without making data, models and code publicly available and usable by other scientists does little to help future scientists attempting the same analyses and less to uncover biases. Authors can only report on biases they already know about, and without the data, models and code, other scientists will be unable to discover issues post hoc.

Table 1 | Proposed reproducibility standards

	Bronze	Silver	Gold
Data published and downloadable	x	x	x
Models published and downloadable	x	x	x
Source code published and downloadable	x	x	x
Dependencies set up in a single command		x	x
Key analysis details recorded		x	x
Analysis components set to deterministic		x	x
Entire analysis reproducible with a single command			x

life sciences community should adopt norms and standards that underlie reproducible machine learning research.

The menu

While many regard the computational reproducibility of a work as a binary property, we prefer to think of it on a sliding scale that reflects the time needed to reproduce. Published works fall somewhere on this scale, which is bookended by 'forever', for a completely irreproducible work, and 'zero', for a work where one can automatically repeat the entire analysis with a single keystroke. As in many cases it is difficult to impose a single standard that divides work into 'reproducible' and 'irreproducible', we instead propose a

dependencies of the analysis can be downloaded and installed in a single command; (2) key details for reproducing the work are documented, including the order in which to run the analysis scripts, the operating system used and system resource requirements; and (3) all random components in the analysis are set to be deterministic. The silver standard is a midway point between minimal availability and full automation. Works that meet this standard will take much less time to reproduce than ones only meeting the bronze standard.

3. Gold standard. The work meets the silver standard, and the authors make the analysis reproducible with a single

Gold

The gold standard for reproducibility requires the entire analysis to be reproducible with a single command. Achieving this goal requires authors to automate all steps of their analysis, including downloading data, pre-processing data, training models, producing output tables, and generating and annotating figures. Full automation stands in addition to tracking dependencies and making their data and code available. In short, by meeting the gold standard, authors make the burden of reproducing their work as small as possible.



Determining the Weight of a Petabyte of Cat Images

Wolfgang Mauere
Technical University of Applied Science Regensburg
Siemens AG, Corporate Research
Regensburg/Munich, Germany
wolfgang.mauere@othr.de

Stefanie Scherzinger
University of Passau
Passau, Germany
stefanie.scherzinger@uni-passau.de

ABSTRACT

We address the question of how to measure the weight of a petabyte worth of random noise in metric units. Quisque consectetur, in suscipit mauris a dolor pellentesque consectetur. Mauris convallis neque non erat. In lacina. Pellentesque leo eros, sagittis quis, fermentum quis, tincidunt ut, sapien. Maecenas sem. Curabitur eros odio, interdum eu, feugiat eu, porta ac, nisl. Curabitur nunc. Etiam fermentum convallis velit. Pellentesque laoreet lacus. Quisque sed elit. Nam quis tellus. Aliquam tellus arcu, adipiscing non, tincidunt eleifend, adipiscing quis, augue. Vivamus elementum placerat enim. Suspendisse ut tortor. Integer faucibus adipiscing felis. Aenean consectetur mattis lectus. Morbi malesuada faucibus dolor. Nam lacus. Etiam arcu libero, malesuada vitae, aliquam vitae, blandit tristique, nisl.

ACM Reference Format:

Wolfgang Mauere and Stefanie Scherzinger. 2021. Determining the Weight of a Petabyte of Cat Images. In *Proceedings of Super Elastic Petabyte Processing Conference (SEPP'21)*. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/nnnnnn.nnnnnnn>

1 INTRODUCTION

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2 RESULTS

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SEPP'21, August 23–25, Athens, Greece

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<https://doi.org/10.1145/nnnnnn.nnnnnnn>

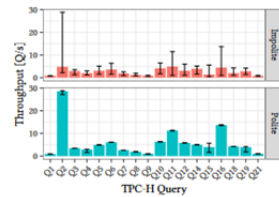


Figure 1: Scatter plot of measurements that show the influence of global warming on Mie scattering of polaritons against storage arrays.

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SEPP'21, August 23–25, Athens, Greece

Wolfgang Mauere and Stefanie Scherzinger

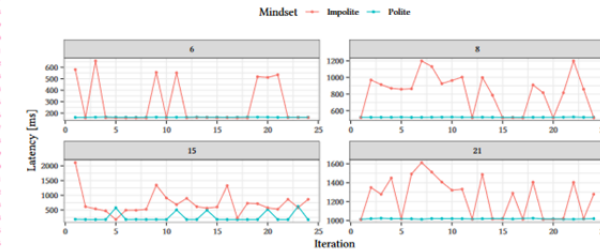


Figure 2: Scatter plot of measurements that show the influence of global warming on Mie scattering of polaritons against storage arrays.

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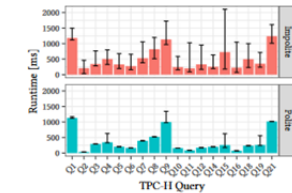


Figure 3: Scatter plot of measurements that show the influence of global warming on Mie scattering of polaritons against storage arrays.

3 CONCLUSIONS

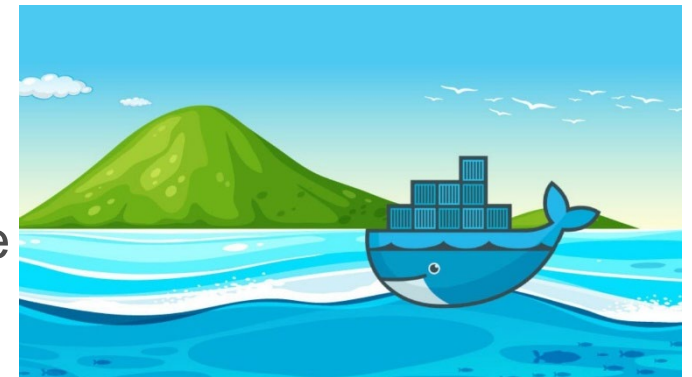
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Be independent of updates to external components, because...

- Base system details change (package versions, etc.)
- System runtime configuration (beyond distro and kernel versions) changes
- GitHub repositories disappear
- Projects move between hosts (Sourceforge, GitHub, ...)
- External software is no longer maintained, download links disappear
- ...

Goal: Build self-contained, complete environments

Be ready to build your stuff even when you are trapped on an island without internet access, or 20 years after all the repositories have gone



<https://stackify.com/docker-image-vs-container-everything-you-need-to-know/>